Stakeholders Communication Planning in a Decision Support System applied to Integrated Water Resource Management

Tatiana Tavares [1]Carlos Gamba [1]Abraham Yu [1]Ricardo Hirata [2]
[1] Technological Research Institute (SPRSF/CIMA/IPT)
[2] Groundwater Research Center, University of São Paulo (CEPAS|USP)

Water resources conflicts around the world will get worse with the increased demand and supply reduction resulting from global climate change. In this scenario, the integrated water resources management (IWRM) makes possible to rationalize the use of water and reduce water insecurity, as it makes compatible the quality and quantity of water, meeting demands, land use planning and the involvement of stakeholders, within certain legal frameworks.

The Decision Support System assists in the dialogue between the various government institutions that participate in the water resources management, users and civil society, as well as in the application of solutions that meet multiple variables and actors.

The inclusion of the human dimension in the water resources management is fundamental, as it affects directly its governance. Scientific uncertainties and the natural systems complexity require the inclusion of the perspectives and concerns of stakeholders, which will influence to make the best decision, prioritizing management options that adjust to available scientific knowledge and social concerns, making the process participatory. Another aspect is how stakeholders will react to possible frustrations of their interests in the construction of a water resources management plan.

An effective stakeholder involvement in the decision-making process is critical in an environment where water security is more than 30% dependent on private wells. Experience in Brazil has shown that command and control solutions (top-down approach) have not been efficient and end up inducing a scenario where 60-70% of wells are irregular or clandestine.

Considering that complex problems in the natural resources management are better solved when there is stakeholder engagement in a processes that involve dialogue, learning and action, the present work seeks to define communication strategies and appropriate approaches to build a trust relationship with stakeholders in two watersheds in Brazil that have historically been suffering from water crises, and where a hydroeconomic model will be built to help decision makers.

This work is part of a larger project (FAPESP Proc. 2020/15434-0), which, in addition to dealing with the management and governance of water resources, seeks to increase water availability using management alternatives, nature based solutions (managed aquifer recharge and phytoremediation) and engineering (contaminated aquifers treatment and riverbank filtration).

It will be carried out with the survey, consolidation and processing of local stakeholders data, searching for define who, when, at what level, and how they should be involved in the decision process. Based on this definition, specific strategies will be adopted to approach and obtain the engagement of all actors, through workshops and dynamics with specific social sectors.

A detailed analysis of the stakeholder groups, during the decision making process, will make possible to improve the understanding of perceptions and engagement processes throughout the construction of the local hydroeconomic model. It is also expected to examine in more detail how stakeholder involvement influences the model results, and how they can be improved throughout this process.